

Jesus E. Vazquez

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Appointments

Johns Hopkins University

POSTDOCTORAL FELLOW

Baltimore, MD

August 2025 - Present

- Johns Hopkins Provost Postdoctoral Fellow in the Department of Biostatistics.
- Developing federated learning methods for incomplete and missing data, with applications to pulmonary health.
- **Advisor:** [Dr. Elizabeth A. Stuart](#).

Education

University of North Carolina at Chapel Hill

PH.D. IN BIOSTATISTICS (2025)

Chapel Hill, NC

August 2019 - August 2025

M.S. IN BIOSTATISTICS (2023)

- Dissertation: Advancing Huntington Disease Modeling: Robust and Efficient Solutions for Right-Censored Covariates.
- **Research Advisor:** [Dr. Tanya P. Garcia](#).

University of New Mexico

B.S. IN STATISTICS, MINOR IN MATHEMATICS (GPA: 3.8/4.0)

Albuquerque, NM

August 2015 - May 2019

B.A. IN ECONOMICS, MINOR IN STATISTICS (GPA: 3.8/4.0)

- Undergraduate honors thesis: Impact of household waste management on health: A case study of Siddharthanagar.
- **Research Advisors:** [Dr. Fares Qeadan](#) and [Dr. Alok Bohara](#).

STUDY ABROAD: RUPANDEHI, NEPAL

December 2017 - January 2018

- Identified risk factors for waterborne diseases using principal component analysis and logistic regression using survey data from the Rupandehi District, Nepal.
- Presented undergraduate honors thesis on the impact of waste management on public health to the mayor and other community leaders.

Grants & Fellowships

Johns Hopkins Provost Postdoctoral Fellowship

JOHNS HOPKINS UNIVERSITY

Baltimore, MD

August 2025 - Present

- One of 4 fellows selected university-wide through a competitive grant-based application to the Johns Hopkins Provost's Office.

R01 Diversity Administrative Supplement

NATIONAL INSTITUTE OF NEUROLOGICAL DISORDERS AND STROKE

Chapel Hill, NC

January 2024 - June 2025

- Wrote a successful Diversity Administrative Supplement application to *Developing a robust and efficient strategy for censored covariates to improve clinical trial design for neurodegenerative diseases* (Tanya Garcia, PI) 1R01NS131225-01.

T32 Environmental Sciences Training Grant

NATIONAL INSTITUTE OF ENVIRONMENTAL HEALTH SCIENCES

Chapel Hill, NC

August 2020 - May 2023

- Three-year predoctoral fellow dedicated to develop innovative methods for addressing challenges in public health and environmental sciences

Doctoral Merit Assistantship for Study in Biostatistics

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Chapel Hill, NC

Fall 2019 - Spring 2020

- Nominated by academic program and selected through a university-wide competition of the most promising future scholars, policymakers, and innovators.

Ronald E. McNair Scholars Program

UNIVERSITY OF NEW MEXICO, DEPARTMENT OF ECONOMICS

Albuquerque, NM

August 2017 - May 2019

- Investigated the importance of pollution and waste management practices in the health of community members of Bhairahawa, Rupandehi, Nepal.

El Puente Undergraduate Research Fellowship

UNIVERSITY OF NEW MEXICO, DEPARTMENT OF ECONOMICS

Albuquerque, NM

August 2016 - April 2017

- Formulated research proposal to investigate how Corporate Social Performance (CSP) affects financial and market performance of companies in the southwest.

Honors & Awards

3rd Place, 3-Minute Presentation Competition

JOHNS HOPKINS POSTDOCTORAL ASSOCIATION CONFERENCE

- Presented *Higher educational attainment in Huntington disease families: evidence from the Enroll-HD study*.

Baltimore, MD

June 2026

Travel Award

SOCIETY FOR ADVANCEMENT OF CHICANOS/HISPANICS AND NATIVE AMERICANS IN SCIENCE (SACNAS)

San Juan, PR

October 2022

1st Place, Poster Competition

UNC DEPARTMENT OF BIostatISTICS 70TH ANNIVERSARY CONFERENCE

- Presented *Validity of the global physical activity questionnaire in the Hispanic Community Health Study/Study of Latinos (HCHS/SOL)*.

Chapel Hill, NC

November 2019

Papers

† DENOTES SENIOR (LAST) AUTHORSHIP.

UNDER REVIEW

5. **Vazquez, J.E.**; Sheng, Y.; Akulian, J.; Hochberg, C.; Iwashyna, T.J.; Stuart, E.A.; Tong, J. **Federated Learning with Missing Data: A Weighted Approach with Variance Correction**. *Statistics in Medicine*. (expected 2026). [arXiv](#)
4. **Vazquez, J.E.**; Ma, Y.; Marder, K.; Garcia, T.P. **Robust Estimation under Outcome Dependent Right Censoring in Huntington Disease: Estimators for Low and High Censoring Rates**. *Statistics in Medicine*. (expected 2026). [arXiv](#)
3. Park, M.; **Vazquez, J.E.**; Qaqish, B.; Jolly, P.S. **CO₂ Laser Assisted Photodynamic Therapy as a Treatment for Basal Cell Carcinoma**. *Journal of Skin Cancer*. (expected 2026).
2. Choi, H.J.; **Vazquez, J.E.**; Qaqish, B.; Bedawi, E.O.; Rahman, N.M.; Yarmus, L.; Akulian, J.; Retrolysis and IPOG Study Groups. **Intrapleural Enzyme Therapy Failure in Pleural Infection: Distinct Factors Associated with Surgery and Mortality in an International Multi-centre Cohort**. *CHEST*. (expected 2026).
1. Diddams, M.J.; **Vazquez, J.E.**; Bedawi, E.; Rahman, N.M.; Yarmus, L.; Qaqish, B.; Akulian, J.A.; Retrolysis and IPOG Study Groups. **Predicting Treatment Failure with Combination Intrapleural Enzyme Therapy for Complicated Pleural Space Infections: The SPARTAN Score**. (expected 2026). [SSRN](#)

PUBLISHED PEER-REVIEWED

7. Sanderson, K.; **Vazquez, J.E.**; Flythe, J.E.; Askenazi, D.; Kosorok, M.R.; Lu, M.; Makker, K.; Laughon, M.; Kistler, C.; Fry, R.C.; O'Shea, T.M. **Perinatal predictors of abnormal adolescent cardiovascular-kidney health after extremely preterm birth: A machine learning analysis of the Extremely Low Gestational Age Newborn-Environmental Child Health Outcomes (ELGAN-ECHO) cohort**. *Kidney International Reports*. 2026. [DOI](#)
6. **Vazquez, J.E.**; Ashner, M.C.; Ma, Y.; Marder, K.; Garcia, T.P. **How Estimators Change When Adapted from the Missing Covariate Problem to the Right-Censored Covariate Problem**. *International Statistical Review*. Accepted May 12, 2026. [DOI](#)
5. **Vazquez, J.E.**; Lin, D.; Mendizabal, A.; Ogilvie, A.C.; Stuart, E.A.; Garcia, T.P. **Higher educational attainment in Huntington disease families: evidence from the Enroll-HD study**. *Orphanet Journal of Rare Diseases*. 2026. [DOI](#)
4. Boerwinkle, V.; Nowlene, M.; **Vazquez, J.E.**; et al. **Resting State fMRI seizure onset localization meta-analysis: Comparing rs-fMRI to other seizure localizing modalities and surgical outcomes**. *Frontiers in Neuroimaging*. 2024. [DOI](#)
3. **Vazquez, J.E.**; Sotres-Alvarez, D.; Carlson, J.; Talavera, G.; Gallo, L.; Castañeda, S.; Evenson, K. **Concurrent validity of the global physical activity questionnaire to accelerometry in Hispanic/Latino adults: The Hispanic Community Health Study/Study of Latinos (HCHS/SOL)**. *Journal of Science and Medicine in Sport*. 2024. [DOI](#)
2. Lotspeich, S.C.; Ashner, M.C.; **Vazquez, J.E.**; Richardson, B.D.; Grosser, K.F.; Bodek, B.E.; Garcia, T.P. **Making sense of censored covariates: statistical methods for studies of Huntington's disease**. *Annual Review of Statistics and Its Application*. 2024. [PMID](#)
1. Rahman, M.M.; Bohara, A.; **Vazquez, J.E.**† **Geospatial analysis of health risks and solid waste management behaviour**. *Journal of Environmental Economics and Policy*. 2021. [DOI](#)

IN PROGRESS

13. **Vazquez, J.E.**; Mendizabal, A.; Ogilvie, A.C.; Nguyen, T.; Garcia, T.P.; Stuart, E.A. **Maternal Predominance Among Affected Parents in Huntington Disease: Evidence from the Enroll-HD Study**.
12. Raman, M.; Ma, Y.; Marder, K.; Garcia, T.P.; **Vazquez, J.E.**† **Multiply-robust inverse probability weighting estimation for right-censored covariates**.
11. **Vazquez, J.E.**; Ringlein, G.; Bang, J.; Stuart, E.A.; Nguyen, T. **Causal inference with a right-censored marker of disease progression: Adjusting for latent severity in Huntington disease**.
10. Rahman, M.; **Vazquez, J.E.**; Lin, D.; Cross, S.; Krishnan, A.; Lotspeich, S.C.; Garcia, T.P. **Multi-state time-to-event modeling of Huntington Disease stage progression**.

9. Lin, D.; Ma, Y.; Garcia, T.P.; **Vazquez, J.E.** † **Identifiability of Regression Models with a Right-Censored Covariate: Extending Canonical Causal Diagrams.**
8. Grosser, K.; Ma, Y.; **Vazquez, J.E.**; Rahman, M.; Lotspeich, S.C.; Garcia, T.P. **Nonparametric Conditional Mean Imputation in Linear Models of Huntington Disease.**
7. **Vazquez, J.E.**; Ma, Y.; Marder, K.; Garcia, T.P. **Longitudinal regression models with a right-censored covariate.**
6. Ochoa-Morales, A.; **Vazquez, J.E.**; Jara-Prado, A.; Yescas-Gómez, P.; Guerrero-Camacho, J.; Cruz-López, F.; Orozco-Pla, M.; Gómez-Pérez, C.A.; Fernandez-Ruiz, J.; Ramírez-García, M.A. **Clinical, Molecular, and Survival Characteristics of Huntington's Disease in Mexico: A 30-Year Cohort from the National Referral Center.**
5. **Vazquez, J.E.**; Nguyen, T.; Ogilvie, A.; Bang, J.; Garcia, T.P.; Stuart, E.A. **Target Trial Emulation of Antidepressant Initiation and Depression Outcomes in Huntington Disease: A Clone-Censor-Weight Approach with Grace Periods.**
4. Alexander-Savino, C.V.; **Vazquez, J.E.**; Qaqish, B.; Culton, D.A. **A retrospective chart review on social determinants of health in pemphigus patients at a tertiary care referral center from 2014–2022 in the United States.**
3. Diddams, M.D.; **Vazquez, J.E.**; et al. **The IET Score: A 30- and 90-day mortality prediction model for patients presenting with complicated pleural infection and treated with intrapleural enzyme therapy.**
2. Pizzuto, M.; Menozzi, S.E.; **Vazquez, J.E.**; Qaqish, B. **Detection of multiple respiratory viruses associated with mortality and severity of illness among children with COVID-19.**
1. **Vazquez, J.E.**; Lotspeich, S.C.; Ma, Y.; Garcia, T.P. **Robustness of parametric maximum likelihood estimation for handling right-censored covariates in generalized linear models.**

PAPERS COMPLETED (NOT PUBLISHED)

4. Adatsi, G.; Hiba, F.; Togo, E.; Young, A.; Hane, A.A.; **Vazquez, J.E.**; et al. **"It Won't Break You": A survival guide for Black, Asian, and minority ethnic graduate students of color in their first year post-George Floyd.** *University of North Carolina at Chapel Hill.* 2024.
3. Qeadan, F.; **Vazquez, J.E.**; Layne, L.; English, K. **Changes in contact prevalence of health conditions from electronic health records after implementation of ICD-10-CM.** Albuquerque Area Southwest Tribal Epidemiology Center. 2019.
2. **Vazquez, J.E.**; Qeadan, F.; Carlson, A. **Electronic medical record-based analysis of optimal screening ages for cerebral aneurysms in adults with Autosomal Dominant Polycystic Kidney Disease (ADPKD).** Albuquerque Area Southwest Tribal Epidemiology Center. 2019.
1. **Vazquez, J.E.**; Yannakopoulos, A.; Johnson, K.; Mancuso, C.; Krishnan, A. **Learning numerical representations of biomedical concepts from 28 million abstracts using neural networks.** Department of Computational Mathematics Science & Engineering, Michigan State University. 2018.

PUBLISHED ABSTRACTS

5. Sanderson, K.; **Vazquez, J.E.**; Makker, K.; Defreitas, M.J.; South, A.M.; Isaac, J.; Aschner, J.L.; Reynolds, M.L.; Kucirka, L.M.; Reidy, K.J.; et al. **Association Between Maternal Preeclampsia and Abnormal Estimated Glomerular Filtration Rate in Adolescent Offspring Born Extremely Preterm.** *Journal of the American Society of Nephrology.* 2025; 36(10S):10–1681. [DOI](#)
4. **Vazquez, J.E.**; Ashner, M.; Ma, Y.; Marder, K.; Garcia, T.P. **F014 Handling censored covariates in Huntington disease studies.** *Journal of Neurology, Neurosurgery & Psychiatry* (BMJ Publishing Group). 2024. [DOI](#)
3. Choi, H.J.; **Vazquez, J.E.**; Qaqish, B.; Akulian, J.A. **Failure of intrapleural enzyme therapy: a multicenter retrospective study.** *CHEST.* 2024; 166(4):A3693. [DOI](#)
2. Diddams, M.; **Vazquez, J.E.**; Qaqish, B.F.; Bedawi, E.; Rahman, N.; Yarmus, L.B.; Akulian, J. **Validation of the RAPID Score for Complicated Pleural Infection Treated With Intrapleural Enzyme Therapy.** *American Thoracic Society 2023 International Conference.* 2023.
1. Diddams, M.; **Vazquez, J.E.**; Qaqish, B.F.; Bedawi, E.; Rahman, N.; Yarmus, L.B.; Akulian, J. **A 90-day Mortality Prediction Model for Patients Presenting With Pleural Infection and Treated With Intrapleural Enzyme Therapy.** *American Thoracic Society 2023 International Conference.* 2023, A4556. [DOI](#)

Presentations

* (INVITED GUEST SPEAKER)

GUEST LECTURES

2. Johns Hopkins Bloomberg School of Public Health, Current Topics in Biostatistics Research (first-year student seminar); **Research overview and career path in biostatistics.** March 2026. *
1. Wake Forest University, STA779: Applied Survival Analysis; **Joint modeling of survival and longitudinal data.** November 2024. *

ORAL PRESENTATIONS

13. **Vazquez, J.E.**; Lin, D.; Mendizabal, A.; Ogilvie, A.C.; Stuart, E.A.; Garcia, T.P.; **Higher educational attainment in Huntington disease families: evidence from the Enroll-HD study.** Johns Hopkins Postdoctoral Association Conference. Baltimore, MD. June 2026. (3rd place, 3-Minute Presentation Competition)
12. **Vazquez, J.E.**; Ma, Y.; Marder, K.; Garcia, T.P.; **Robust estimation under outcome dependent right censoring in Huntington disease: Estimators for low and high censoring rates.** International Conference on Computational Statistics (COMPSTAT). Athens, GR. August 2026.
11. **Vazquez, J.E.**; Shen, Y.; Akulian, J.; Hochberg, C.; Iwashyna, T.J.; Tong, J.; Stuart, E.A.; **Federated learning with incomplete data: Applications to preterm kidney health.** International Biometric Conference (IBC). Seoul, KR. July 2026.

10. **Vazquez, J.E.;** Shen, Y.; Akulian, J.; Hochberg, C.; Iwashyna, T.J.; Tong, J.; Stuart, E.A.; **Federated learning with incomplete data: Applications to preterm kidney health.** Eastern North American Region (ENAR) Spring Meeting. 2026.
9. **Vazquez, J.E.;** Ma, Y.; Marder, K.; Garcia, T.P.; **Robust estimation under outcome dependent right censoring in Huntington disease: Estimators for low and high censoring rates.** Johns Hopkins Causal Inference Working Group. Baltimore, MD. November 2025. *
8. Choi, H.J.; **Vazquez, J.E.;** Qaqish, B.; Akulian, J.; et al.; **Assessment of factors associated with the failure of intrapleural enzyme therapy in the treatment of pleural infection.** American College of Chest Physicians (CHEST) Annual Meeting. Boston, MA. October 2025.
7. **Vazquez, J.E.;** Ashner, M.C.; Ma, Y.; Marder, K.; Garcia, T.P.; **Handling censored covariates in Huntington disease studies.** EHDN & Enroll-HD Conference. Strasbourg, FR. September 2024.
6. **Vazquez, J.E.;** Ashner, M.C.; Ma, Y.; Marder, K.; Garcia, T.P.; **The missing link: Establishing the parallels between right-censored and missing covariates.** International Society for Clinical Biostatistics. Thessaloniki, GR. July 2024. *
5. **Vazquez, J.E.;** Ashner, M.C.; Ma, Y.; Marder, K.; Garcia, T.P.; **Evaluating the robustness and efficiency of estimators under informative covariate censoring.** Banff International Research Station. Banff, CA. February 2024. * ([Video](#))
4. **Vazquez, J.E.;** Ma, Y.; Marder, K.; Garcia, T.P.; **Using biostatistics estimators to model the progression of symptoms for Huntington disease in astronomy.** iid2022: Statistical Methods for Event Data Illuminating the Dynamic Universe. Huntsville, AL. November 2022.
3. **Vazquez, J.E.;** Qeadan, F.; Carlson, A.; **Autosomal dominant polycystic kidney disease (ADPKD) and cerebral aneurysms.** Clinical & Translational Investigators Program Seminar, UNM School of Medicine. Albuquerque, NM. February 2019.
2. **Vazquez, J.E.;** Yannakopoulos, A.; Johnson, K.; Mancuso, C.; Krishnan, A.; **Learning numerical representations of biomedical concepts from 28 million abstracts.**

October 2018
October 2018
July 2018

Ronald McNair Research Scholars Conference, University of Maryland
Ronald McNair Research Scholars Conference, University of New Mexico
iCER Symposium, Michigan State University

Baltimore, MD
Albuquerque, NM
East Lansing, MI

1. **Vazquez, J.E.;** Bohara, A.; **Household waste management effects on health and behavior: A case study of Siddharthanagar.** Pratiman-Neema Memorial Health Institute. Rupandehi District, NEP. December 2017.

POSTER PRESENTATIONS

8. **Vazquez, J.E.;** Ma, Y.; Marder, K.; Garcia, T.P.; **Robust estimation under outcome dependent right censoring in Huntington disease: Estimators for low and high censoring rates.** Workshop on Statistics & Data Science. Bogotá, CO. December 2025.
7. **Vazquez, J.E.;** Ma, Y.; Marder, K.; Garcia, T.P.; **Estimators for longitudinal mixed models to account for right-censored covariates.**

March 2023
December 2022

Eastern North American Regional Conference, ENAR
International Mathematical Statistics Conference, IMS

Nashville, TN
Florence, IT

6. **Vazquez, J.E.;** Sotres-Alvarez, D.; Carlson, J.; Talavera, G.; Gallo, L.; Castañeda, S.; Evenson, K.; **Validity of the global physical activity questionnaire in the Hispanic Community Health Study/Study of Latinos (HCHS/SOL).** University of North Carolina at Chapel Hill. Chapel Hill, NC. November 2019. (1st place, Poster Competition)
5. **Vazquez, J.E.;** Collman, B.; Morris, R.M.; **Biased childhood asthma mortality rates in the US: A vector autoregressive approach.** University of New Mexico. Albuquerque, NM. May 2019.
4. **Vazquez, J.E.;** Qeadan, F.; Carlson, A.; **Autosomal dominant polycystic kidney disease (ADPKD) and cerebral aneurysms.** University of New Mexico School of Medicine. Albuquerque, NM. May 2018.
3. **Vazquez, J.E.;** Riera Villa, N.; Anderson, N.; **NaCl effects on hydroponically grown kale and mustard greens.** University of Minnesota. Minneapolis, MN. December 2017.
2. **Vazquez, J.E.;** Bohara, A.; **Household waste management effects on health and behavior: A case study of Siddharthanagar.** Undergraduate Research Opportunity Conference, University of New Mexico. Albuquerque, NM. April 2017.
1. **Vazquez, J.E.;** Bohara, A.; **Is arsenic contamination the real problem, or is it just one of the factors?** University of New Mexico. Albuquerque, NM. May 2017.

Teaching

GR (GRADUATE LEVEL), UG (UNDERGRADUATE LEVEL)

BIOS H500: Intro to Biostatistics (UG)

Chapel Hill, NC

TEACHING ASSISTANT-BIOSTATISTICS DEPARTMENT, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Fall 2022

- Held office hours that guided students in sampling design, descriptive statistics, probability, confidence intervals, ANOVA, nonparametric methods, regression, survival analysis.

BIOS 664: Applied Survey Data Analysis (GR)

Chapel Hill, NC

TEACHING ASSISTANT- BIOSTATISTICS DEPARTMENT, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Spring 2022

- Fundamental principles and methods of sampling populations, with emphasis on simple random, stratified, and cluster sampling.

BIOS 667: Applied Longitudinal Data Analysis (GR)

Chapel Hill, NC

TEACHING ASSISTANT- BIOSTATISTICS DEPARTMENT, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Fall 2021

- Guided students to help understand linear and generalized linear models marginal and mixed effects models, methods of estimation (MLE, REML, GEE), variance components, hypothesis testing and confidence intervals, prediction of random effects, regression diagnostics, sample size calculation.

BIOS 663: Intermediate Linear Models (GR)

Chapel Hill, NC

TEACHING ASSISTANT- BIOSTATISTICS DEPARTMENT, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Spring 2021

- Led recitations and created HW answer keys for matrix-based treatment of regression, one-way and two-way ANOVA, and ANCOVA, with applications in the health sciences.

BIOM 505: Biostatistics 1 & 2 (GR)

Albuquerque, NM

TEACHING ASSISTANT- SCHOOL OF MEDICINE, THE UNIVERSITY OF NEW MEXICO

Fall 2018 & Spring 2019

- Supported and held office hours for medical students without a statistical background in regression, ANOVA, survival analysis and power analysis.

Econ 395: Economic Issues (UG)

Albuquerque, NM

TEACHING ASSISTANT - ECONOMICS DEPARTMENT, UNIVERSITY OF NEW MEXICO

Fall 2017

- Assisted students in using STATA for data analysis in public health, focusing on multiple linear regression, probability concepts, and survey design applications.

UNIV101: Transitional Studies (UG)

Albuquerque, NM

TEACHING ASSISTANT, UNIVERSITY OF NEW MEXICO

Spring 2017

- Supported the course by recommending readings to first-generation college students, grading assignments, and helping students coordinate community service projects.

Leadership & Service Roles

Planning Committee Member

Baltimore, MD

JOHNS HOPKINS POSTDOCTORAL ASSOCIATION

May 2026 - Present

- Served on the planning committee for postdoctoral community events at Johns Hopkins University.
- Secured many of the social events for Postdoc Appreciation Week and Postdoc Appreciation Month.

Methods for Incomplete Data Lab Facilitator

Chapel Hill, NC

BIOSTATISTICS DEPARTMENT, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

August 2024 - May 2025

- Schedule meetings, send out calendar invites, and booked meeting rooms.
- Organize talks and arrange snacks schedule.

Social Media Committee Member Chair

Chapel Hill, NC

EASTERN NORTH AMERICAN REGION (ENAR) INTERNATIONAL BIOMETRIC SOCIETY

August 2024 - August 2025

- Announced awards and winners, industry guidance updates and regulatory updates, among other material.
- Created content and shared announcements from new journal issues of Biometrics and JABES; Providing timely commentaries and other news from the Biometric Bulletin.

Interdepartmental Student Liason

Chapel Hill, NC

UNC BIOSTATISTICS STUDENT ASSOCIATION

August 2019 - May 2022

- Ensured the Biostatistics Department was regularly updated on events from the Statistics & Operations Research Department (STOR) to strengthen interdepartmental partnerships.

Departmental Student Representative

Chapel Hill, NC

UNC GILLINGS SCHOOL OF PUBLIC HEALTH STUDENT GOVERNMENT ASSOCIATION

August 2020 - May 2022

- Addressed concerns of Biostatistics students and organized fundraising events to support student activities within the public health school.

Diversity, Equity, and Inclusion (DEI)

Fostering Excellence in Biostatistics Workshop

Baltimore, MD

EASTERN NORTH AMERICAN REGION (ENAR) INTERNATIONAL BIOMETRIC SOCIETY

September 2024 - Present

- **2026:** Served as Social Media Committee Chair and Organizing Committee Member; chaired the Student Panel and Introduction to Biostatistics Kahoot sessions.
- **2025:** Led a team in reviewing and posting advertising materials as Social Media Committee Chair.
- **2024:** Planning committee member.

Graduate School Mentorship Initiative (GSMI)

CIENTÍFICO LATINO

Baltimore, MD

August 2026 - Present

- Mentor students from backgrounds underrepresented in STEM through their graduate school applications, providing field-specific guidance, feedback on written materials, and mock interviews.

Graduate School Application Reviewer and Mentor Program

MENTORSHIP AND ADVICE FOR PROSPECTIVE STUDENTS, UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Chapel Hill, NC

August 2024 - May 2025

- Assisted underrepresented biostatistics applicants in reviewing their graduate school application materials to optimize their chances of being accepted to the Department of Biostatistics, University of North Carolina at Chapel Hill.

Public Relations Manager & Founding Member

LATINE GRADUATE & PROFESSIONAL STUDENT ASSOCIATION (LGPSA), UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Chapel Hill, NC

April 2022 - April 2023

- Co-founded organization seeking to uphold a welcoming environment for Latine graduate and professional students.
- Responsible for raising awareness and visibility of LGPSA across the UNC-Chapel Hill community through channels such as radio show interviews and organizing panel discussions of graduate students.
- Led fundraising events including designing and selling t-shirts.

VOLUNTEERING

Fall 2024	Maureen Joy Charter School , Spanish interpreter for Hispanic/Latino parents.	Durham, NC
Summer 2023	MESA National Engineering Design Competition , Discussed personal struggles of homelessness and path to higher education in STEM to High School students.	Albuquerque, NM
Spring 2023	Adelante Education Coalition , Led a group of Hispanic/Latinx high school students into various workshops focused on exploring their aspirations beyond professional life.	High Point, NC
Summer 2022	Biostatistics Undergraduate Student Internships , Helped organize symposium for the student-led summer internship program for biostatistics undergraduate students at UNC.	Chapel Hill, NC
Summer 2022	EqualityNC , Helped distribute information of EqualityNC's support LGBTQ programs during the Raleigh Pride 2022.	Raleigh, NC
Summer 2022	Juntos Summer Academy , Presented about personal journey in conducting research in public health and Biostatistics to Latino/Hispanic High School Students. Presented them with opportunities for them to get involved.	Raleigh, NC
Spring 2022	Codepath , Participated as a resume reviewer to increase minority representation in technology.	Chapel Hill, NC

Other Relevant Experience

Graduate Student Intern

SANDIA NATIONAL LABORATORIES

Albuquerque, NM

May 2023 - August 2023

- Led research project estimating and testing the reduction in photovoltaic energy production during the 2018 Camp Wildfire using spatial regression methods based on Markov Random Fields.
- Trained and evaluated semi-parametric and machine learning models for predicting photovoltaic energy production by integrating diverse ocean, atmospheric, and terrain spatial data sources.
- **Research Advisor:** [Dr. Alex Washburne](#)

Data Scientist Intern

GOOGLE

San Francisco, CA

May 2020 - August 2020

- Proposed an Inter-Platform-Reliability metric to measure the reliability of users when answering YouTube surveys. The validity of the metric was then evaluated using a simulation study.
- Recommended safe-hypothesis testing practices for Kappa coefficients under small sample-sizes.

Project Assistant

COLLEGE ASSISTANCE MIGRANT PROGRAM, UNIVERSITY OF NEW MEXICO

Albuquerque, NM

August 2016 - December 2017

- Assisted in planning student activities, collecting data, administering outreach, and evaluating programs for undergraduate students from low-income and farmworker backgrounds.
- Met with local farmworkers and members of the community to organize the 2017 Farmworkers Awareness Week.
- Helped students to coordinate and execute community service projects.

References

JOHNS HOPKINS UNIVERSITY

Dr. Elizabeth A. Stuart, Professor of Biostatistics and Department Chair

POSTDOC ADVISOR (ESTUART@JHU.EDU)

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Dr. Tanya P. Garcia, Associate Professor of Biostatistics

DISSERTATION ADVISOR (TPGARCIA@EMAIL.UNC.EDU)

Dr. Bahjat Qaqish, Professor Emeritus of Biostatistics

RESEARCH SUPERVISOR (QAQISH@BIOS.UNC.EDU)

PENN STATE UNIVERSITY

Dr. Yanyuan Ma, Professor of Statistics

COLLABORATOR (YZM63@PSU.EDU)